

A Novel Approach to Landslide Prediction: Transformer with Positional Encoding and Logistic Regression

Arshdeep Kaur
Computer Science and Engineering
Sardar Beant Singh State University
Gurdaspur, Punjab, India
mearshdeep603@gmail.com

Ajay Kumar Dogra
Computer Science and Engineering
Sardar Beant Singh State University
Gurdaspur, Punjab, India
akdogra@sbssugsp.ac.in

Abstract— The landslide prediction is crucial task for disaster management, and to address socio-economic and environmental consequences. In this paper, a hybrid model is implemented for landslide prediction and achieved overall accuracy of 94.7%. The model composed of Transformer with Positional Encoding and Logistic Regression. It is suitable for making prediction using multivariate spatial-temporal data, including rainfall, slope, land cover, and soil type. Initially, data preprocessing is performed with label encoding, feature scaling, and dimensionality reduction using an auto-encoder. Then, reduced feature set is classified using logistic regression, optimized via grid search. The proposed method achieved an accuracy of 94.70%, with strong weighted precision of 94.95% and recall of 94.70%. Also, Class-wise analysis indicated high performance for dominant classes but lower specificity for minority classes. The results demonstrate the potential of combining advanced feature extraction and optimized classification for accurate landslide prediction.

Keywords— Land Slide Prediction; Transformer; Logistic Regression; Positional Encoding; Machine Learning.

I. INTRODUCTION

Landslides are one of the most devastating natural hazards, often result into severe socio-economic and environmental consequences. The reason being, accurate and timely prediction of landslide occurrences is critical for early warning systems, risk mitigation, and disaster management, particularly in vulnerable terrains exposed to intense rainfall, seismic activity, and human interventions [1]. In contrast to which, traditional geotechnical and statistical methods, often struggle to capture the nonlinear and dynamic relationships among multiple influencing factors such as rainfall, soil type, slope gradient, and land use [2].

According to the reports UNDRR, 2022, landslides cause an estimated 4,800 deaths annually and billions of dollars in economic losses worldwide. Also, increased rainfall intensity due to climate change is projected to raise landslide frequency in many mountainous regions by 30–50% by 2050 [22]. Furthermore, the Indian subcontinent, particularly the Himalayan belt and Western Ghats, is among the world's most landslide-prone regions, with over 12% of its landmass susceptible. As per GSI, 2022, about 12.6% of India's land area is prone to landslides GSI, 2022 [23].

However, recent advancements in Machine Learning (ML) and Deep Learning (DL) have significantly improved the performance of predictive models for landslide susceptibility mapping. While approaches like decision trees, support vector machines, and artificial neural networks have shown promise, many models still face limitations in handling spatial and sequential dependencies within environmental data [3].

To address the limitations, in this paper, a novel hybrid model that integrates the Transformer architecture with positional encoding for feature representation and a logistic regression classifier has been introduced for final prediction. The transformer model can capture long-range dependencies in sequential data without relying on relapse [4]. By embedding positional information into the feature representation, the model can understand spatial and temporal order within the input variables, highly relevant to landslide modelling where geographic and time-related dependencies are critical [5].

Unlike standard deep learning models, the proposed approach avoids overfitting and leverages the powerful attention-based feature encoding of transformers. Then employs logistic regression layer for classification. This model is designed to be lightweight and suitable for deployment in real-time monitoring systems. Otherwise, traditional statistical models often fail to capture nonlinear interactions between environmental variables, making machine learning a vital tool. Only a limited number of regions globally have operational, real-time landslide early warning systems integrating multi-source data [23].

The experimental evaluation, conducted on a real-world landslide data set, demonstrates that the proposed transformer-logistic regression pipeline achieves high accuracy, recall, precision and F1-score, outperforming traditional models and offering a scalable solution for landslide risk prediction [6].

This paper is sectioned into section 1 for Introduction to the research problem and proposed solution; section 2 for Related Work; section 3 for Dataset Description; section 4 for Implementation; section 5 for Results and Discussion; section 6 for Conclusion and Future Scope.

I. RELATED WORK

TABLE 1: COMPARATIVE LITERATURE

Sr. No.	Author	Approach	Algorithm Used	Accuracy Achieved
[1]	Wu, B., Shi, Z., Zheng, H. et al.	Machine Learning	Decision Trees, Random Forest, Adaboost, and Gradient Tree Boosting	AUC of 91.2% with Random Forest
[2]	Nanda, A.M., Lone, F.A. & Ahmed, P.	Machine Learning	Random Forest (RF) and Logistic Regression (LR)	AUC of 92% with Random Forest
[3]	Javidan, N., Kavian, A., Conoscenti, C. et al	Machine Learning	Random Forest (RF), Support Vector Machine (SVM), Boosted Regression Tree (BRT), and Multivariate Adaptive Regression Splines (MARS)	AUC of 99.1% using Multivariate Adaptive Regression Splines (MARS)
[4]	Madhu, D., Nithya, G.K., Sreekala, S. et al.	Machine Learning	Logistic Regression	AUC of 91%
[5]	Li, B., Li, C., Liu, Y. et al.	Deep Learning	DDL-based Collaborative Feature Neural Network (DDL- CFNN)	76% lesser prediction errors
[11]	Li, J., Fan, C., Zhao, K. et al	Deep Learning	LSTM-TCN	AUC of 97.6%
[6]	Wang, Y., Zhou, C., Cao, Y. et al.	Deep Learning	CNN	AUC of 92%
[7]	Saha, S., Barman, A., Saha, A. et al.	Deep Learning	DLNN	AUC of 89.07%
[12]	Li, J., Fan, C., Zhao, K. et al	Computer Vision	Feature vector, Artificial Neural Network (ANN) and Support Vector Machine (SVM) models	AUC of 87.6%
[13]	Wang, Y., Zhou, C., Cao, Y. et al.	Computer Vision	ResNet18, U-Net, and VGG16	AUC of 91% with ResNet
[14]	Qiu, Q., Xie, Z., Zhang, D. et al.	Computer Vision	automated geological Hazard knowledge graph	AUC of 89%
[8]	Şener, A., Ergen, B.	Hybrid	Hybrid Attentional Atrous Convolution	AUC of 97.6%
[9]	Chen, C., Fan, L.	Hybrid	ML and DL models - IGR, REF, PSO, LASSO, HHO	AUC of 93%
[10]	Zhang, Y., Yang, L. et al.	Hybrid	SCC-RFFS-RF	AUC of 91%
[15]	W., Shu, H., Abbas, H. et al.	Hybrid	GAN with logistic regression	AUC of 87%

II. DATASET DESCRIPTION

The data collected for this research work is secondary data collected from an online platform Kaggle.com. The dataset is named as "Regenerated Landslide Risk Dataset", publicly available on Kaggle. It provides a rich set of environmental, geological, and infrastructural features that influence landslide occurrence. The data is composed of different variables such as temperature, humidity, precipitation, soil moisture, and elevation, based on which the occurrence of landslide is predicted, with 5000 records.

III. PROPOSED MODEL OR APPROACH

The proposed approach is composed of transformer, autoencoder, and logistic regression as a classifier. The transformer model can capture long-range dependencies in sequential data without relying on relapse. By embedding positional information into the feature representation, the model can understand spatial and temporal order within the input variables, highly relevant to landslide modelling where geographic and time-related dependencies are critical [12].

Then, an Autoencoder is used, an unsupervised deep learning model that learns a compressed representation of the input data. The network-based Autoencoder is employed for non-linear dimensionality reduction, and reducing model complexity [22].

Later, to establish a baseline predictive model for landslide prediction, Logistic Regression is selected. The model has high interpretability and effectiveness for binary classification tasks. The combination is expected to increase prediction accuracy and produce outcomes that can be explained for emergency and policy planning, with higher accuracy [23].

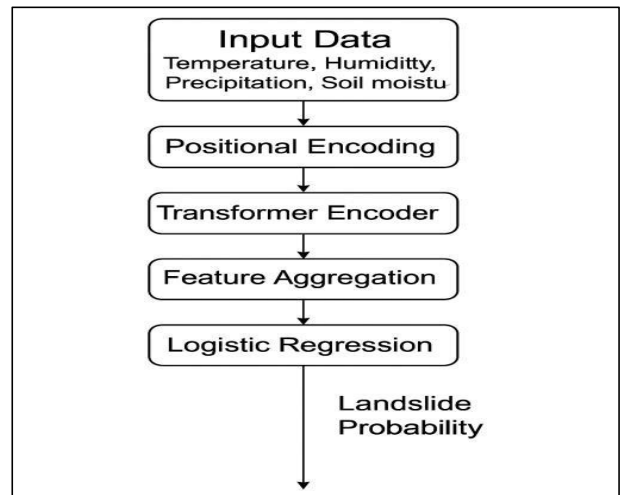


Fig 1 Proposed Method

IV. IMPLEMENTATION

After data collection, the land slide prediction is performed in Python based platform using some important libraries such as pandas, numpy, sklearn, tensorflow.keras, seaborn, and

matplotlib. The libraries are used to perform data loading, pre-processing, predictions, visualization etc.

A. Data Loading and Pre-processing

Initially, the data is loaded from .csv file and read with different parameters such as temperature, humidity, precipitation, soil moisture, and elevation. Then, data is cleaned by dropping all the unnecessary columns, and records with null values. Then, a heat map is generated to find correlation between all the variables considered for predicting landslides [11].

Originally, the correlations between the variables are extremely close to zero, means there is no linear relationship between them. For landslide prediction, precipitation and soil moisture are typically the strongest predictors, with elevation as a structural or geographical factor. Temperature and humidity are secondary but still useful when building multi-factor predictive models, especially in specific climate zones.

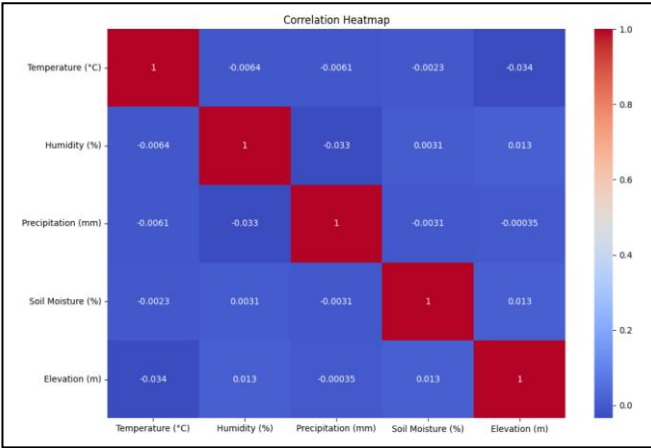


Fig 2. Correlation between different variables

1) Categorical Variable Encoding

Being the part of data pre-processing, then categorical encoding is performed. The dataset contained non-numerical values such as soil classification, land cover type etc. Since machine learning algorithms require numeric input, these categorical variables are converted into numerical form using the Label Encoding technique from the scikit-learn library [15]. Label Encoding assigns a unique integer to each category within a feature. For example, the values {'Clay', 'Silt', 'Sand'} would be encoded as {0, 1, 2} respectively. These encoding preserves category identity while enabling direct integration into the predictive model. This transformation ensures all features are numeric, facilitating correlation analysis and machine learning model training for landslide prediction [16].

2) Feature scaling using Transformer

The dataset's features varied across different ranges. For example, precipitation was measured in millimeters, while soil moisture was in percentage and elevation in meters. Such differences in scale can bias machine learning models. To normalize the feature ranges, Standardization is applied using Standard Scaler.

This method transforms each feature to have a mean of zero and a standard deviation of one, improving numerical stability and ensuring that all variables contribute equally to the

model.fit_transform() computes the mean and standard deviation for each feature from the training data and scales the features accordingly [17].

3) Dimensionality reduction using Autoencoder

After feature scaling, dimensionality is reduced. High-dimensional feature spaces can introduce noise, increase computational complexity, and reduce model generalization due to the curse of dimensionality. To address this, a neural network-based Autoencoder is employed for non-linear dimensionality reduction [18].

An Autoencoder is an unsupervised deep learning model that learns a compressed representation of the input data, while also attempting to reconstruct the original input. Unlike traditional linear methods such as PCA, Autoencoders can capture complex, non-linear relationships between variables, making them suitable for multi-factor environmental datasets such as those used in landslide prediction [22].

B. Data splitting

After standardization, the feature matrix is passed through the trained Autoencoder encoder to obtain a reduced-dimensionality representation of the data [22]. This transformation compressed the original high-dimensional feature space into 20 latent features, in the encoding step. The reduced dataset is then partitioned into training and testing subsets using the train-test split technique, where 80% of the data was used for model training and 20% for evaluation.

C. Model Training

Later, to establish a baseline predictive model for landslide prediction, Logistic Regression is selected. The selected model has high interpretability and effectiveness for binary classification tasks [23]. To identify the optimal model configuration, Grid Search Cross-Validation (GridSearchCV) is applied with a 5-fold strategy.

D. Model Evaluation

Then performance of the model is evaluated using classification metrics such as precision, recall, f1-score, and support. The Logistic Regression model, optimized via Grid Search, achieved an overall accuracy of 94.70% on the test dataset. The overall evaluation of model' classification is performed using precision, recall, f1-score, and support. Otherwise, per-class performance of the model is evaluated using accuracy, sensitivity (recall), specificity, and precision.

V. RESULTS AND DISCUSSION

Class 1, representing the majority class, showed excellent performance with 0.98 precision and 0.97 recall, ensuring reliable detection of major events. Minority classes (0 and 2) had moderate precision and recall, indicating a need for improved rare-event detection, while Class 3 scored perfectly due to a single instance.

Macro averages including precision 0.81, recall 0.84 suggest balanced performance across classes, though weighted averages 0.95 reflect dominance of Class 1. High recall for critical events is vital in disaster prediction, and future work should address class imbalance using oversampling or weighting techniques.

Classification Report:				
	precision	recall	f1-score	support
0	0.60	0.64	0.62	14
1	0.98	0.97	0.98	902
2	0.68	0.73	0.71	83
3	1.00	1.00	1.00	1
accuracy			0.95	1000
macro avg	0.81	0.84	0.83	1000
weighted avg	0.95	0.95	0.95	1000

Fig 3. Classification metrics

Later, per-class performance is evaluated to gain deeper insights into the model’s performance beyond aggregate metrics, using per-class accuracy, sensitivity (recall), specificity, and precision. For each class, True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN) are calculated, based on which accuracy, sensitivity (recall), specificity, and precision, are predicted [19].

Per-class evaluation showed Class 1 achieving the highest balance of sensitivity of 0.99 and precision of 0.99, though specificity was moderate of 0.64. Class 2 and Class 3 had strong overall accuracies of 0.956 and 0.949 but lower sensitivities for rare-event detection of 0.81 and 0.97. Class 4 scored perfect across all metrics due to a single correctly classified instance. In this evaluation, the classes are different level of risks relative to landslide occurrence [20].

The model achieved an overall accuracy of 94.70%, indicating strong general performance. Macro-averaged sensitivity of 83.72%, precision of 81.44%, and F1-score of 82.53%, reflect balanced detection across all classes, including minority categories. Weighted metrics such as sensitivity of 94.70%, precision of 94.95%, and F1-score of 94.81%, highlight excellent performance on the dominant class, suggesting the model is highly reliable for common landslide events but could improve rare-event detection [21].

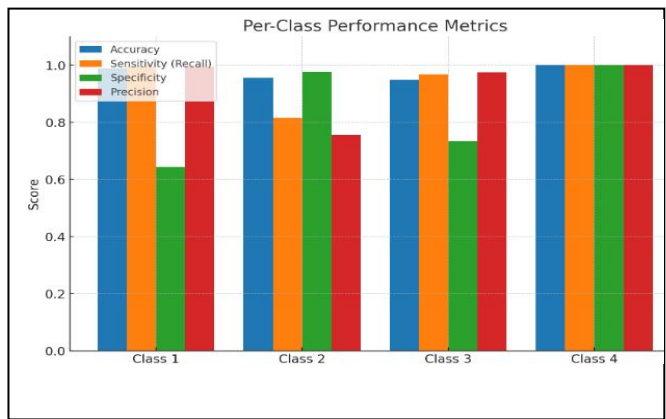


Fig 4. Per-Class Performance

In Figure 4, the confusion matrix reveals strong classification performance, particularly for Class 1, with 876 correct predictions and minimal misclassification, reflecting the model’s bias toward the dominant class. Class 0 and Class 2 exhibit moderate confusion, with misclassifications occurring between these categories, suggesting overlapping feature patterns. Class 3 achieved perfect prediction but contains only one instance, limiting generalizability. The results highlight the model’s robustness for frequent classes while indicating challenges in distinguishing minority classes.

This imbalance, common in landslide datasets, affects macro-averaged metrics despite high overall accuracy 94.70%.

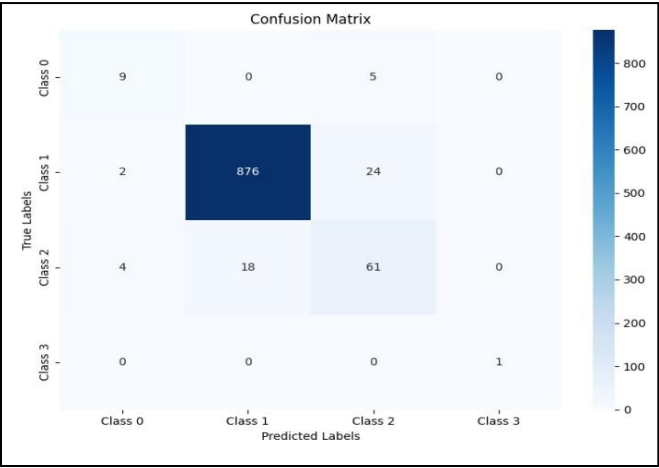


Fig 5. confusion matrix

Overall, Classes 0 and 2, suggesting overlapping feature characteristics. High weighted metrics reflect the dominance of Class 1, while macro scores reveal the effect of imbalance.

VI. FUTURE SCOPE

In future, the proposed landslide prediction framework can be modified by integrating Transformer encoder architectures for improved spatial-temporal feature extraction, enabling the model to better capture long-range dependencies in environmental patterns. Also, incorporating additional real- time data sources, such as satellite imagery and IoT-based rainfall sensors, could further enhance prediction accuracy and timeliness. Furthermore, the framework can also be extended into multi-hazard **prediction systems** and deployed as a cloud-based early warning service for disaster management authorities, improving scalability, accessibility, and response efficiency.

VII. CONCLUSION

In conclusion, the model is obtained as simple but high performance to predict landslides based on different parameters. An overall accuracy of 94.70%, demonstrating high effectiveness in landslide risk prediction. Also, Class-wise analysis shows excellent performance for the majority Class 1 with 98% precision and 97% recall, while minority classes including Class 0 and Class 2 had lower specificity, indicating some misclassification. Along with this, the confusion matrix reveals most errors occurred between Classes 0 and 2, suggesting overlapping feature characteristics.

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